



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

Seat No.							
----------	--	--	--	--	--	--	--

Institute: (0701)SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Programme: (070126) BACHELOR OF TECHNOLOGY (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

Batch: 2021-25,2022-26

Semester: VI

Course: Reinforcement Learning

Course Code: 0701260603

Date: 21/05/2025

Maximum Marks: 45

Day: Wednesday

Time: 13:30 - 15:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. All questions are compulsory.
2. Write your seat number in word and figures in the answer sheet provided.
3. The use of non-programmable calculator is allowed.
4. Please note that the content of CO, BL and PI is for administrative purpose.
5. This question paper consists of 2 printed sheets.

Part A

		Marks	CO/BL/PI
Q.1	a. Describe how state and action stochasticity is modelled in a RL process using appropriate equations.	3	1/2/1.2.1
	b. A person can be in two moods: $S = \{\text{Happy}, \text{Sad}\}$ The transitions are: If they are Happy today, there is an 80% chance they stay Happy , and a 20% chance they become Sad . If they are Sad today, there's a 50% chance they stay Sad , and a 50% chance they become Happy . Draw a state transition diagram and write the state transition matrix.	4	2/2/2.5.2
	c. Define the concept of model in RL with the use of appropriate equations.	2	3/2/2.5.2
	d. Justify: why is it not possible to solve model free control problems using state value function $V(s)$?	2	4/5/3.5.1

- | | | | |
|----|--|---|-----------|
| e. | Draw the state-action diagram to show the difference between MC, TD and DL. | 3 | 5/3/4.4.1 |
| f. | Is target policy in Q-Learning greedy or ϵ -greedy in nature and why? | 2 | 6/4/4.4.1 |

Part B

- | | | | |
|-----|---|---|-----------|
| Q.2 | Show bellman equation of optimality in the form of $Q(S,A)$ with the help of a suitable diagram. | 3 | 1/3/4.4.2 |
| Q.3 | Analyse the value iteration method with appropriate equations and algorithm. | 5 | 2/4/2.5.2 |
| Q.4 | <p>The Critic has two states A and B following 8 episodes:</p> <ul style="list-style-type: none"> • $s_a, r = 0, s_b, r = 0, \text{END}$ • $s_b, r = 1, \text{END}$ • $s_b, r = 1, \text{END}$ • $s_b, r = 1, \text{END}$ • $s_b, r = 1, \text{END}$ • $s_b, r = 1, \text{END}$ • $s_b, r = 1, \text{END}$ • $s_b, r = 1, \text{END}$ • $s_b, r = 0, \text{END}$ <p>Draw a diagram showing the above episodes and transitions. Find $V(A)$ and $V(B)$ using both MC and TD methods.</p> | 6 | 3/3/3.5.1 |
| Q.5 | Formulate n-step TD return with help of equations and suitable diagram. | 5 | 4/6/2.5.1 |
- OR**
- | | | | |
|-----|---|---|-----------|
| Q.5 | Write a note on Dyna-Q algorithm. | 5 | 4/6/2.5.1 |
| Q.6 | Which function approximators are used in RL? What is the need to use function approximators in RL models? | 5 | 5/4/2.8.1 |
| Q.7 | Define epsilon-greedy policy iteration process with the help of an appropriate equation and diagram. | 5 | 6/5/3.5.1 |



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category – I by UGC

Seat No.

--	--	--	--	--

Institute: (0701)SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Programme: (070126) BACHELOR OF TECHNOLOGY (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

Batch: 2021-25,2022-26

Semester: VI

Course: Reinforcement Learning

Course Code: 0701260603

Date: 12/11/2025

Maximum Marks: 45

Day: Wednesday

Time: 13:30 - 15:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. All questions are compulsory.
2. Write your seat number in word and figures in the answer sheet provided.
3. The use of non-programmable calculator is allowed.
4. Please note that the content of CO, BL and PI is for administrative purpose.
5. This question paper consists of 2 printed sheets.

	Part A	Marks	CO/BL/PI
Q.1	a. What is the need and motivation to use Reinforcement Learning in some applications?	3	1/2/1.2.1
	b. You are modeling the weather. The states are: $S = \{\text{Sunny, Rainy}\}$ The transition probabilities are: If it is Sunny today, there's a 70% chance it will be Sunny tomorrow and 30% chance it will be Rainy . If it is Rainy today, there's a 60% chance it will be Rainy tomorrow and 40% chance it will be Sunny . Draw a state transition diagram and write the state transition matrix.	3	2/2/2.5.2
	c. Explain: Bellman Expectation Equation in the form of $Q(S,A)$	2	3/2/2.5.2
	d. If the state transition probabilities and reward is unknown, can you use Dynamic Programming to solve such RL problem?	2	4/2/3.5.1

- | | | | |
|----|---|---|-----------|
| e. | With help of appropriate equations explain the difference between TD off-policy learning and TD on-policy Learning. | 3 | 5/2/4.4.1 |
| f. | Differentiate between exploration and exploitation with a simple day-to-day example. | 2 | 6/4/4.4.1 |

Part B

- | | | | |
|------------|--|---|-----------|
| Q.2 | Distinguish between:
a. Full observability and partial observability
b. Episodic and continuous tasks | 4 | 1/4/4.4.2 |
| Q.3 | $A + 3 \rightarrow A + 2 \rightarrow B - 4 \rightarrow A + 4 \rightarrow B - 3 \rightarrow \text{Terminate}$
$B - 2 \rightarrow A + 3 \rightarrow B - 3 \rightarrow \text{Terminate}$
Calculate $V(A)$ and $V(B)$ using the First visit MC method. | 6 | 2/3/3.5.1 |
| Q.4 | Discuss the use of incremental mean in model-free methods with the help of equations. | 6 | 3/3/2.5.2 |
| Q.5 | Explain MC prediction method with help of an algorithm. | 4 | 4/5/2.5.1 |
| Q.6 | Describe in detail Deep Q Learning with help of an algorithm. | 5 | 5/2/2.8.1 |

OR

- | | | | |
|------------|---|---|-----------|
| Q.6 | Discuss in detail: use of experience relay in Deep Q Networks | 5 | 5/2/2.8.1 |
| Q.7 | How is exploration and exploitation achieved in Model free control methods? Explain with help of equations. | 5 | 6/2/3.5.1 |



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

Seat No.

--	--	--	--	--	--

Institute: (0701)SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Programme: (070126) BACHELOR OF TECHNOLOGY (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

Batch: 2021-25

Semester: VI

Course: Reinforcement Learning

Course Code: 0701260603

Date: 29/11/2024

Maximum Marks: 45

Day: Friday

Time: 13:30 - 15:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. All questions are compulsory.
2. Write your seat number in words and figures in the answer sheet provided.
3. The use of non-programmable calculators is allowed.
4. Please note that the content of CO, BL and PI is for administrative purpose.
5. This question paper consists of 3 printed sheets.

PART A

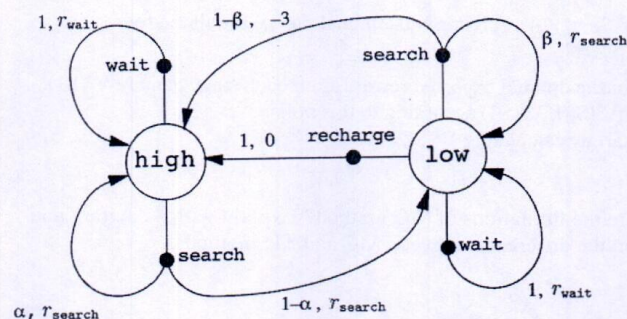
Q.1 a. Explain the concept of Markov Decision process with a diagram.

Marks CO/BL/PI

3 1/2/1.2.1

b. For the given model, write the transition probability matrix.

3 2/2/2.5.2



c. Explain: Bellman Optimality Equation

2 3/2/2.5.2

d. What is meant by Model based methods?

2 4/2/3.5.1



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

Seat No.

--	--	--	--	--	--

Institute: (0701)SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Programme: (070126) BACHELOR OF TECHNOLOGY (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

Batch: 2021-25

Semester: VI

Course: Reinforcement Learning

Course Code: 0701260603

Date: 29/05/2024

Maximum Marks: 45

Day: Wednesday

Time: 13:30 - 15:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

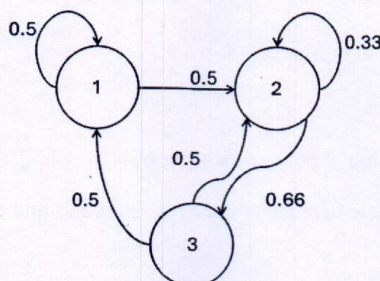
Instructions:

1. All questions are compulsory.
2. Write your seat number in words and figures in the answer sheet provided.
3. The use of non-programmable calculators is allowed.
4. Please note that the content of CO, BL and PI is for administrative purpose.
5. This question paper consists of 3 printed sheets.

PART A

Marks CO/BL/PI

- Q.1
- a. Write a note on Markov Rule, Markov Chain, Markov Reward Process. 3 1/2/1.2.1
 - b. For the given model, write the transition probability matrix. 3 2/2/2.5.2



- c. Define: State Value function and action value function with equations. 2 3/2/2.5.2
- d. Justify: why is it not possible to solve model free control problems using state value function $V(s)$? What is the solution for this? 4 4/5/3.5.1
- e. Draw the unified view of Reinforcement Learning. 2 5/2/4.4.1

- f. Distinguish between exploration vs exploitation with a simple day-to-day example.

2 6/4/4.4:1

PART B

- Q.2** Discuss in detail:
a. state value function
b. bellman equation of optimality

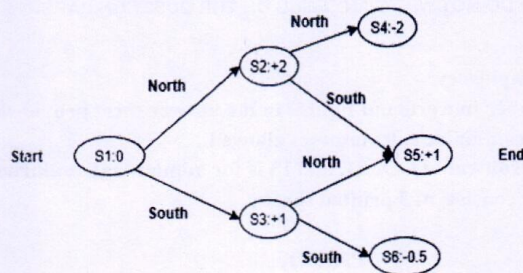
5 1/2/4.4.2

- Q.3** For the given example:

S1: Start state, S2: State of landing in gold Island, S3: State of landing in silver Island, S4: State of landing in Bermuda Triangle Island, S5: State of landing in destination Island, S6: State of landing in prison Island.

6 2/3/3.5.1

The rewards obtained in a particular state are mentioned next to each state.



There are two actions: south and north.

The state transition probabilities matrices are as shown for action 'south' and 'north'.

$$T[A(North)] = \begin{bmatrix} S1 & S2 & S3 & S4 & S5 & S6 \\ S1 & 0 & 0.8 & 0.2 & 0 & 0 & 0 \\ S2 & 0 & 0 & 0 & 0.8 & 0.2 & 0 \\ S3 & 0 & 0 & 0 & 0 & 0.8 & 0.2 \\ S4 & 0 & 0 & 0 & 0 & 0 & 0 \\ S5 & 0 & 0 & 0 & 0 & 0 & 0 \\ S6 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad T[A(South)] = \begin{bmatrix} S1 & S2 & S3 & S4 & S5 & S6 \\ S1 & 0 & 0.2 & 0.8 & 0 & 0 & 0 \\ S2 & 0 & 0 & 0 & 0.2 & 0.8 & 0 \\ S3 & 0 & 0 & 0 & 0 & 0.2 & 0.8 \\ S4 & 0 & 0 & 0 & 0 & 0 & 0 \\ S5 & 0 & 0 & 0 & 0 & 0 & 0 \\ S6 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Start with a random initial policy of $\pi = [N, N, N, N, N, N]$.

Carry out one complete iteration of policy iteration and find the improved policy.

- Q.4** What are the differences between MC and TD methods of model free RL.

4 3/4/2.5.2

- Q.5** Explain in detail: SARSA with the help of appropriate equations and algorithm.

4 4/2/2.5.1

Q. 6

Explain with diagrams and equations, difference between Q learning and Deep Q Learning.

5

5/4/2.8.1

OR

Which different function approximators are used in RL? What is the need to use function approximators RL models?

5

5/4/2.8.1

Q. 7

Justify why is it important to balance between exploration and exploitation? Explain how this is achieved in MC control.

5

6/5/3.5.1



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

Seat No.

--	--	--	--	--	--

Institute: (0701)SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Programme: (070126) BACHELOR OF TECHNOLOGY (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

Batch: 2021-25

Semester: VI

Course: Reinforcement Learning

Course Code: 0701260603

Date: 29/05/2024

Maximum Marks: 45

Day: Wednesday

Time: 13:30 - 15:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

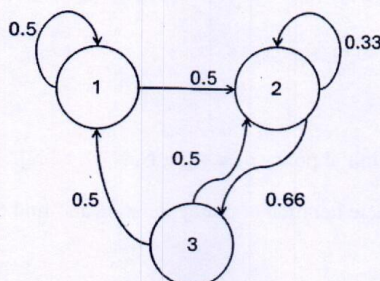
Instructions:

1. All questions are compulsory.
2. Write your seat number in words and figures in the answer sheet provided.
3. The use of non-programmable calculators is allowed.
4. Please note that the content of CO, BL and PI is for administrative purpose.
5. This question paper consists of 3 printed sheets.

PART A

Marks CO/BL/PI

- Q.1
- a. Write a note on Markov Rule, Markov Chain, Markov Reward Process. 3 1/2/1.2.1
 - b. For the given model, write the transition probability matrix. 3 2/2/2.5.2



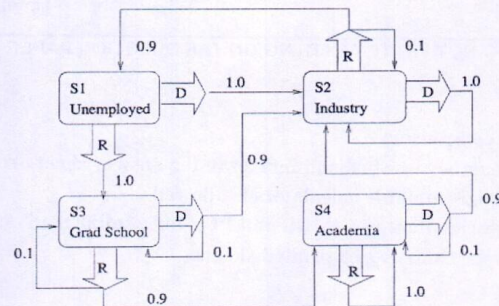
- c. Define: State Value function and action value function with equations. 2 3/2/2.5.2
- d. Justify: why is it not possible to solve model free control problems using state value function $V(s)$? What is the solution for this? 4 4/5/3.5.1
- e. Draw the unified view of Reinforcement Learning. 2 5/2/4.4.1

- e. Draw a simple diagram explaining the difference between Q-learning and deep Q learning. 2 5/2/4.4.1
- f. Differentiate between exploration vs exploitation with a simple day-to-day example. 2 6/4/4.4.1

PART B

- Q.2 Distinguish between:
a. Deterministic vs stochastic policy
b. Continuous vs discrete system 5 1/4/4.4.2

- Q.3 Consider the MDP below. The state transition probabilities are as shown from action to state. There are two actions out of each state. D for development and R for research. 6 2/3/3.5.1



The immediate rewards of starting in each state are:

REWARD			
S1	S2	S3	S4
0	100	0	10

For $\gamma = 0.9$, $\pi^*(s) = D$ represents optimal policy for all the four states S1, S2, S3, S4.

Solve for the optimal expected reward from each state, namely $V^*(S1)$, $V^*(S2)$, $V^*(S3)$, $V^*(S4)$ according to this policy.

Hint: Start by calculating $V^*(S2)$.

- Q.4 What are the limitations of MC method? Explain with a diagram and equation the difference between MC and TD methods. 6 3/2/2.5.2

OR

Write a note on integration of learning and planning. What are the various methods for this. 6 3/2/2.5.2

- Q.5 Differentiate between: on-policy learning vs off-policy learning. 4 4/4/2.5.1

Q. 6 Write a detailed note on Deep Q Learning with help of an algorithm. 5 5/2/2.8.1

Q. 7 Explain how the exploration and exploitation both is achieved in Model free control methods. 5 6/2/3.5.1
